
Catalyst 9300-M Datasheet

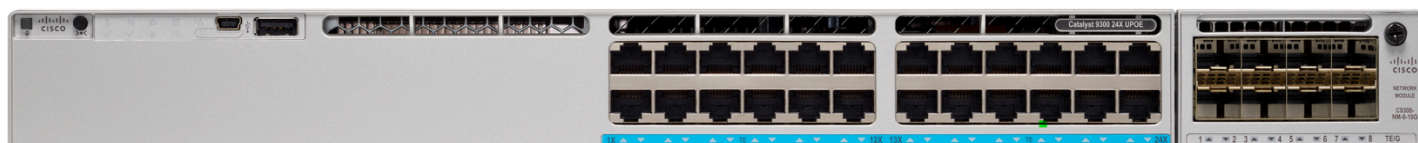
Overview

The Catalyst 9300-M addresses the most demanding enterprise applications by combining the simplicity of the Meraki dashboard with powerful switching hardware. To satisfy high-bandwidth applications and the deployment of high-speed wifi access points, the Catalyst 9300-M provides multigigabit ports, 480G stacking, and modular 10/25/40G uplinks. The Catalyst 9300-M delivers resiliency with fast stack convergence and StackPower. The Catalyst 9300-M provides Adaptive Policy using an over-the-wire tag which segments traffic into security groups to deliver scalable security. The Catalyst 9300-M is integrated under the Meraki dashboard to provide a simply powerful solution to the most demanding wired access applications.

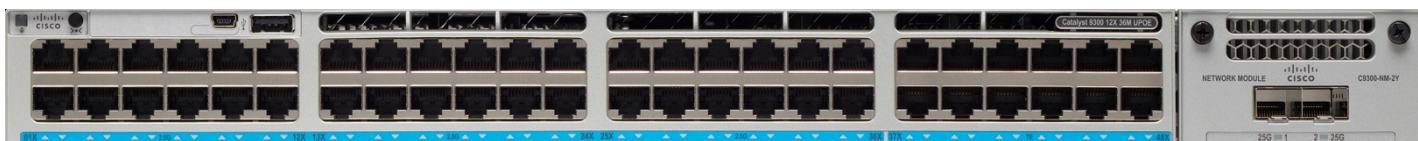
Adaptive Policy provides simple & scalable security policies to segment traffic using Security Groups. Security Groups are created in the dashboard using natural language such as "IOT device" & "Guest." The security policy intent (e.g., Permit or Deny) is then simply provisioned between Security Groups which results in the segmentation of each group's traffic. By making security policy management intuitive and scalable relative to legacy IP-address based Access Control Lists, Adaptive Policy empowers operators to confidently secure their network traffic independent of future network changes.

By pooling & distributing power across C9300-Ms using a series of StackPower cables, StackPower provides simple and resilient power distribution across the stack.

C9300-24UX-M



C9300-48UXM-M



A Catalyst 9300 switch can also be migrated to a Meraki-managed persona, where it would function as its Catalyst Meraki 9300 equivalent. For details on the migration process, please refer to our [Getting started: Cisco Catalyst 9300 with Meraki Dashboard](#) guide.

Models

Number of Ports	Model	Description
Fiber	C9300-24S-M	24-port SFP switch
	C9300-48S-M	48-port SFP switch
24 Ports	C9300-24T-M	24-port GbE switch
	C9300-24P-M	24-port GbE PoE+ switch
	C9300-24U-M	24-port GbE UPoE switch
	C9300-24UX-M	24-port mGbE UPoE switch
48 Ports	C9300-48T-M	48-port GbE switch
	C9300-48P-M	48-port GbE PoE+ switch; ENERGY STAR®? certified
	C9300-48U-M	48-port GbE UPoE switch
	C9300-48UXM-M	36-port 2.5GbE + 12-port mGbE UPoE switch; ENERGY STAR®? certified
	C9300-48UN-M	48-port 5GbE UPOE switch

Features

Catalyst 9300/L/X-M Licensing

The Catalyst 9300-M license applies to all c9300 hardware platforms (c9300 / c9300L / c9300X) excluding the MS390. Catalyst 9300-M license structure includes two feature tiers: Enterprise and Advanced. As with all Meraki managed switches, every Catalyst 9300-M license is available in 1, 3, 5, 7, and 10 year terms.

Catalyst 9300X-M License Structure	12/24-Port	48-port
Advanced Features	LIC-C9300-24A	LIC-C9300-48A
Enterprise Features	LIC-C9300-24E	LIC-C9300-48E

The features available with advanced licensing are:

- [Adaptive policy](#)
- [Netflow v10 \(IPFIX\) and Encrypted Traffic Analytics](#)



In the Co-term licensing model (most existing Organizations), an Organization must either have all Catalyst 9300/L/X-M Enterprise or all Catalyst 9300/L/X-M Advanced licenses - they cannot be mixed. In the Per-device licensing model, a mix of Enterprise and Advanced can be added to a single



Organization, but certain features may require all devices in a Network to have Advanced licenses, e.g. Adaptive Policy.



Additional Information: [MS390s](#) existing in the same co-term Organization will require the **same** license edition (either Enterprise or Advanced) as the Catalyst 9300/L/X-M. License types cannot be mixed and will result in an incompatible license error.

In a co-term license Organization - MS130, MS150, MS390, C9300 require the **same license tier** (either Enterprise or Advanced) across the entire Organization.

For example, if an organization has existing MS390/C9300/MS150/MS130 with enterprise licenses, they cannot add C9300's with advanced licenses in that organization - they must be added with enterprise licenses.

If the organization has existing MS390/C9300/MS150/MS130 with advanced licenses, they cannot add C9300's with enterprise licenses in that organization - they must be added with advanced licenses.

For more information on licensing, refer to [Meraki Licensing Models article](#).

[Subscription licensing](#) is also available for the Catalyst 9300-M switches

Technical Breakdown

Interfaces Specifications

Model/Persona	Interfaces	Uplink (SFP28, QSFP+)	120G Hardware Stack Port	Dedicated Management Interface	PoE/ UPoE Capabilities
C9300-24S-M	24 x 1G SFP	Modular	2	1	n/a
C9300-24T-M	24 x 1GbE RJ45	Modular	2	1	n/a
C9300-24P-M	24 x 1GbE RJ45	Modular	2	1	PoE
C9300-24U-M	24 x 1GbE RJ45	Modular	2	1	UPoE
C9300-24UX-M	24 x 100M/1G/2.5G/5G/ 10G RJ45	Modular	2	1	UPoE
C9300-48S-M	48 x 1G SFP	Modular	2	1	n/a
C9300-48T-M	48 x 1GbE RJ45	Modular	2	1	n/a
C9300-48P-M	48 x 1GbE RJ45	Modular	2	1	PoE
C9300-48U-M	48 x 1GbE RJ45	Modular	2	1	UPoE
C9300-48UXM-M	36 x 100M/1G/2.5G +	Modular	2	1	UPoE

Model/Persona	Interfaces	Uplink (SFP28, QSFP+)	120G Hardware Stack Port	Dedicated Management Interface	PoE/ UPoE Capabilities
	12 x 100M/1G/2.5G/5G/10G				
C9300-48UN-M	48 x 100M/1G/2.5G/5G	Modular	2	1	UPoE

i All models are available with modular uplinks that have been listed under the [Accessories](#) list. For supported SFP modules please refer the [SFP Datasheet](#).

i **Cabling Best Practices for Multi-Gigabit operations:** While Category-5e cables can support multigigabit data rates upto 2.5/5 Gbps, external factors such as noise, alien crosstalk coupled with longer cable/cable bundle lengths can impede reliable link operation. Noise can originate from cable bundling, RFI, cable movement, lightning, power surges and other transient event. It is recommended to use Category-6a cabling for reliable multigigabit operations as it mitigates alien crosstalk by design.

Physical Specifications

Model	Dimensions (h x w x d) W/ Default Power Supply	Weight	Mount Type	Hot Swap Fans	Operating Temperature	Humidity
C9300-24S-M	1.73" x 17.5" x 17.7" (4.3 x 44.4 x 44.9cm)	16.84 lb (7.64 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-24T-M	1.73" x 17.5" x 17.7" (4.4 x 44.5 x 44.9cm)	16.03 lb (7.27 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-24P-M	1.73" x 17.5" x 17.7" (4.4 x 44.5 x 44.9cm)	16.33 lb (7.4 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-24U-M	1.73" x 17.5" x 19.2" (4.4 x 44.5 x 44.8cm)	16.63 lb (7.54 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-24UX-M	1.73" x 17.5" x 20.2" (4.4 x 44.5 x 51.3cm)	18.18 lb (8.25 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%

Model	Dimensions (h x w x d) W/ Default Power Supply	Weight	Mount Type	Hot Swap Fans	Operating Temperature	Humidity
C9300-48S-M	1.73" x 17.5" x 19.2" (4.3 x 44.4 x 44.9cm)	17.32 lb (7.86kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-48T-M	1.73" x 17.5" x 17.7" (4.4 x 44.5 x 44.9cm)	16.43 lb (7.45 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-48P-M	1.73" x 17.5" x 17.7" (4.4 x 44.5 x 44.9cm)	16.73 lb (7.59 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-48U-M	1.73" x 17.5" x 19.2" (4.4 x 44.5 x 48.8cm)	17.03 lb (7.72 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-48UXM-M	1.73" x 17.5" x 22.2" (4.4 x 44.5 x 56.4cm)	20.50 lb (9.34 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%
C9300-48UN-M	1.73" x 17.5" x 22.2" (4.4 x 44.5 x 56.4cm)	20.05 lb (9.09 kg)	1U Rack Mount	Yes, 3x	-5°C to 45°C	5 to 90%

Performance

Model	Switching Capacity	Switching Capacity with Stacking	Forwarding rate
C9300-24S-M	208 Gbps	480 Gbps	154.76 Mpps
C9300-24T-M	208 Gbps	480 Gbps	154.76 Mpps
C9300-24P-M	208 Gbps	480 Gbps	154.76 Mpps
C9300-24U-M	208 Gbps	480 Gbps	154.76 Mpps
C9300-24UX-M	640 Gbps	480 Gbps	476.19 Mpps
C9300-48S-M	256 Gbps	480 Gbps	190.48 Mpps
C9300-48T-M	256 Gbps	480 Gbps	190.48 Mpps
C9300-48P-M	256 Gbps	480 Gbps	190.48 Mpps

Model	Switching Capacity	Switching Capacity with Stacking	Forwarding rate
C9300-48U-M	256 Gbps	480 Gbps	190.48 Mpps
C9300-48UXM-M	580 Gbps	480 Gbps	431.54 Mpps
C9300-48UN-M	640 Gbps	480 Gbps	476.19 Mpps

Power Supply and Specifications

Model	Default Power Supply	Hot Swap Power Supply	Available PoE W/ Primary PS	Max Available PoE W/ Secondary PS**	Power Load (idle/max)
C9300-24S-M	PWR-C1-715WAC-P-M	Yes, Dual	n/a	n/a	84.60 / 124.30 W
C9300-24T-M	PWR-C1-350WAC-P-M	Yes, Dual	n/a	n/a	79.2 / 99 W
C9300-24P-M	PWR-C1-715WAC-P-M	Yes, Dual	445W	720W	84.1 / 554.4 W
C9300-24U-M	PWR-C1-1100WAC-P-M	Yes, Dual	830W	1440W	85.4 / 990.3 W
C9300-24UX-M	PWR-C1-1100WAC-P-M	Yes, Dual	560W	1440W	162.7 / 809.9 W
C9300-48S-M	PWR-C1-715WAC-P-M	Yes, Dual	n/a	n/a	88.90 / 158.30 W
C9300-48T-M	PWR-C1-350WAC-P-M	Yes, Dual	n/a	n/a	83.9 / 109.9 W
C9300-48P-M	PWR-C1-715WAC-P-M	Yes, Dual	437W	1152W	92.6 / 555 W
C9300-48U-M	PWR-C1-1100WAC-P-M	Yes, Dual	822W	1800W	145 / 844.9 W
C9300-48UXM-M	PWR-C1-1100WAC-P-M	Yes, Dual	490W	1590W	218.5 / 785.5 W
C9300-48UN-M	PWR-C1-1100WAC-P-M	Yes, Dual	645W	1745W	157.9 / 843.8 W



** The PoE values are provided considering the secondary PS to be the default power supply of the respective model.

Note: For Japan, the recommendation is to deploy the switch with 200V+ outlets. The 1100WAC power supplies will not operate on 100V and



requires input power range 115V to 240 VAC, 50 to 60 Hz. Please reference [Table 13. AC Power Cords](#) for the appropriate 250VAC power cable.

PSU configuration and PoE budgets

Model	Primary PSU	PoE	With 350W Secondary PSU	With 715W Secondary PSU	With 1100W Secondary PSU
C9300-24P-M	PWR-C1-715WAC-P-M	445W	720W ²	720W ²	720W ²
	PWR-C1-1100WAC-P-M ¹	720W ²	720W ²	720W ²	720W ²
C9300-24U-M	PWR-C1-1100WAC-P-M	830W	1180W ²	1440W ²	1440W ²
C9300-24UX-M	PWR-C1-1100WAC-P-M	560W	910W	1275W	1440W ²
C9300-48P-M	PWR-C1-715WAC-P-M	437W	787W	1152W	1440W ²
	PWR-C1-1100WAC-P-M ¹	822W	1172W	1440W ²	1440W ²
C9300-48U-M	PWR-C1-1100WAC-P-M	822W	1172W	1537W	1800W ³
C9300-48UXM-M	PWR-C1-1100WAC-P-M	490W	840W	1205W	1590W
C9300-48UN-M	PWR-C1-1100WAC-P-M	645W	995W	1360W	1745W

¹ Using upgraded power supply option.

² Limited by port density and per-port PoE capacity.

³ Limited by design.

Package Contents

In addition to the MS switch, each shipment includes the following.

- 1 x default power supply unit for the switch model (please refer to the [power supply and specifications](#) section)
- Rack mount brackets (rack screws not included)
- 3 x Pre-installed fans.



Note: Contact your Meraki partner or reseller to confirm if a power cord is included with your order.

Optional Accessories

Accessory	Description	Supported Models
PWR-C1-350WAC-P-M	350W AC Power Supply	C9300, C9300L, C9300X
PWR-C1-715WAC-P-M	715W AC Power Supply	C9300, C9300L, C9300X
PWR-C1-1100WAC-P-M	1100W AC Power Supply	C9300, C9300L, C9300X
PWR-C1-1900WAC-P-M	1900W AC Power Supply (230V)**	C9300, C9300X
C9300X-NM-4C	4 x 40/100G QSFP28 Uplink Module	C9300X-48HX, C9300X-48TX and C9300X-24Y
C9300X-NM-8Y-M	8 x 25G SFP28 Uplink Module	C9300X
C9300X-NM-2C-M	2 x 40/100G QSFP28 Uplink Module	C9300X
C9300-NM-2Q-M	2 x 40G QSFP Uplink Module	C9300
C9300-NM-8X-M	8 x 1G/10G SFP+ Uplink Module	C9300
C9300-NM-2Y-M*	2 x 25G/10G/1G SFP28 Uplink Module	C9300
C9300-NM-4M	4 x 100M/1G/2.5G/5G/10G RJ-45 Uplink Module	C9300
C9300-NM-4G	4 x 1G SFP Uplink Module	C9300
STACK-T1-50CM-M	Catalyst Meraki 480G Stacking Cable, 0.5 Meter	C9300, C9300X
STACK-T1-1M-M	Catalyst Meraki 480G Stacking Cable, 1 Meter	C9300, C9300X
STACK-T1-3M-M	Catalyst Meraki 480G Stacking Cable, 3 Meter	C9300, C9300X
CAB-SPWR-30CM-M	Catalyst 9300-M 30CM StackPower Cable	C9300, C9300X
CAB-SPWR-150CM-M	Catalyst 9300-M 150CM StackPower Cable	C9300, C9300X
FAN-T2-M	System Fan	C9300, C9300L, C9300X
4PT-KIT-T2-M	Catalyst 9300-M Rack Mount Kit	C9300, C9300L, C9300X



***Note:** The C9300-NM-2Y-M module is supported on C9300-24S/48S with firmware CS16.7. Support for the other C9300 switch models require CS16.8

***Note:** The **SFP28/QSFP28** modules will be orderable March 2024 with software support in CS16.7. **25Gbps Dashboard support** is slated for **CS17** firmware. 25Gbps optics can be used in CS16.8, but should be considered 'Beta'.

****NOTE:** The **PWR-C1-1900WAC-P-M** provides 1900 W output power when the nominal input voltage is 230 V. If the nominal input voltage is 115 V, then the output power is limited to 1500 W. The power supply **does not work with standard 125VAC power cords**(common with 1100W and lower power supplies), you require a C21 power cable like the North American(**C9K-PWR-CAB-AC-US**). Also a good option is the universal C20-C21 jumper cable(**C9K-PWR-CAB-AC-BL**), suitable for Rack PDUs . See [Table 2](#) for all suitable worldwide power cord options. Please note there are no



-M versions of these cables, they must be ordered with the Cisco SKU.



Cisco Optics Compatibility

Many of the Cisco Optics that are **compatible*** with the native Catalyst Switch platform (ex. C9300-24S) are also compatible with the corresponding Cloud Managed/Monitored Catalyst Switch Platform (ex.: C9300-24S-M) including the MS390.

- This includes: 1/10/25/40/100G SFP/QSFP *fixed wavelength* fiber and copper transceivers,
- Please reference the [TMGmatrix](#) for Cisco's official optic-to-device compatibility list. Click this [LINK](#) for a filtered list of all optics for C9300/C9300L/C9300X platforms
- These compatible optics are covered under the existing Cisco Warranty terms available [here](#)
- Meraki support is able to assist with troubleshooting, identifying, and root-causing the issues where the Cisco Optics are being used with their cloud managed/monitored catalyst Switch
- However, if the root cause indicates that the customer is eligible for an RMA, support will continue to follow established procedures for Cisco branded optics, which may include collaborating with Cisco TAC

Note: Validation of dashboard reporting of Cisco Optics is ongoing. In the event Meraki Support identifies an issue with a Cisco Optic, a support case may be filed for further investigation.

Other non-M accessories

Other accessories such as fan spares, power stacking cables, stackwise cables and power supplies are **compatible*** between non-M and -M models.

Some Examples :

- STACK-T1-50CM is compatible with STACK-T1-50CM-M
- CAB-SPWR-30CM is compatible with CAB-SPWR-30CM-M
- FAN-T2 is compatible with FAN-T2-M
- PWR-C1-1100WAC-P is compatible with PWR-C1-1100WAC-P-M

* **Compatibility/ Compatible:** The hardware accessory has been determined to operate effectively with the listed switch platforms, and will receive troubleshooting assistance from Cisco Meraki Support in the event of an issue experienced with the hardware. However, in the event of a hardware replacement requirement an additional step may be required to facilitate the RMA which would require collaboration with Cisco TAC under the Cisco warranty terms. Meraki's hardware warranty and return policy and Meraki EOL policy does not apply to Cisco Optics, Cables and accessories in any shape or form.

SFP Modules

The following SFP/Fiber transceivers are supported. For detailed SFP specifications, refer to the [Meraki SFP datasheet](#).

For information about Cisco Optics compatibility and Cisco's [TMGmatrix](#), please click [here](#).

Transceiver Model	Standard	Supported C9300-M platform
1 Gbps		
MA-SFP-1GB-SX	1000BASE-SX over OM1/2/3/4 up to distances of 1KM	C9300, C9300L, C9300X

MA-SFP-1GB-LX10	1000BASE-LX10 over OS1/OS2 up to distances of 10KM	
10 Gbps		
MA-SFP-10GB-SR	10GBASE-SR over OM1/2/3/4 up to distances of 400 M	
MA-SFP-10GB-LR	10GBASE-LR over OS1/2 up to distances of 10 KM	C9300, C9300L, C9300X
MA-SFP-10GB-ER	10GBASE-ER over OS1/2 up to distances of 40 KM	
MA-SFP-10GB-ZR	10GBASE-ZR over OS1/2 up to distances of 80 KM	
MA-SFP-10GB-LRM	10GBASE-LRM over OM1/2/3, OS1/2 up to distances of 300 M	C9300, C9300L
25 Gbps	Please see TMGmatrix for compatibility for SFP28 modules	C9300, C9300L, C9300X
40 Gbps		
MA-QSFP-40G-CSR4	QSFP-MPO12 over OM2/3/4 up to distances of 400 M	C9300, C9300X
MA-QSFP-40G-SR-BD	QSFP Duplex LC over OM2/3/4 up to distances of 150 M	
100 Gbps		
MA-QSFP-100G-LR4	100GBASE-LR4 over OS1/2 up to distances of 10 KM	C9300X
MA-QSFP-100G-SR4	100GBASE-SR4 over OM3/4/5 up to distances of 100 M	
Direct Attach		
MA-CBL-TA-1M	10GSFP+Cu Passive Twinax	C9300, C9300L, C9300X
MA-CBL-TA-3M		



NOTE: Catalyst 9300X switches do not support LRM modules.

Regulations and Compliance

Electromagnetic Compatibility

Certifications

FCC Part 15 (CFR 47) Class A, ICES-003 Class A, CISPR22 Class A, CNS13438, EN 300 386 V1.6.1, EN 55022 Class A, EN 61000-3-2, EN61000-3-3, KN 32, TCVN 7189 Class A, EN 55032, CISPR 32 Class A, V-2/2015.04, V-3/2015.04, VCCI-CISPR 32 Class A, CISPR24, EN 300 386 V1.6.1, EN 55024, KN35, TCVN 7317

Safety

CAN/CSA-C22.2 No. 60950-1, UL 60950-1, EN 60950-1, IEC 60950-1, AS/NZS 60950.1

Environmental	Reduction of Hazardous Substances (RoHS)
Warranty	Full lifetime hardware warranty with next-day advanced replacement included

MTBF Rating

Model	MTBF at 25°C
C9300-24S-M	284,130
C9300-24T-M	314,790
C9300-24P-M	299,000
C9300-24U-M	238,410
C9300-24UX-M	214,760
C9300-48S-M	281,920
C9300-48T-M	305,870
C9300-48P-M	277,770
C9300-48U-M	227,410
C9300-48UXM-M	202,160
C9300-48UN-M	198,647

Installation Guide

For instructions on how to install and configure the C9300 series switch, refer the [C9300 Series Installation Guide](#).